

EXPERT TIPS

Recursion

When a function calls itself, this is known as recursion. It's another way of making a loop in your program. In most uses of recursion, the parameters of the function change each time the function is called. In Kaleido-spiral, for example, the size, angle, and position of the circle change whenever the function calls itself.

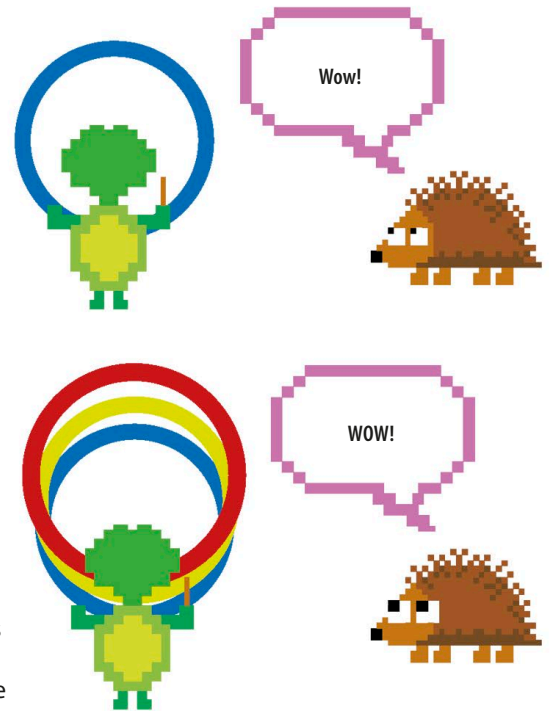


7 Change the colour, increase the size

To create more exciting patterns, make these changes to the code to increase the size of the circle and change its colour. This code uses the `cycle()` function, which takes a list of values as its parameter and returns a special type of list that you can cycle through endlessly using the `next()` function. Run the code again.

6 Test your code

Run the program. You would see the turtle drawing the same circle over and over again. Don't worry – you'll fix that in the next step.



```
import turtle
from itertools import cycle

colors = cycle(['red', 'orange', 'yellow', 'green', 'blue', 'purple'])

def draw_circle(size):
    turtle.pencolor(next(colors))
    turtle.circle(size)
    draw_circle(size + 5)

turtle.bgcolor('black')
turtle.speed('fast')
turtle.pensize(4)
draw_circle(30)
```

Import the `cycle()` function.

This line creates a cycle of the colours in the list.

Use the next colour in the cycle.

Add 5 to the previous circle size.